

X.500 Protocol

The X.500 protocol was first approved in 1988 and then enhanced in 1993 under the auspices of the International Telecommunications Union (ITU). Its purpose was to provide an international standard for directory systems.

Model

The X.500 protocol architecture consists of a Client-Server communicating via the Open Systems Interconnection (OSI) networking model. The Client is called the Directory Service Agent (DUA) and the Server is called the Directory System Agent (DSA). There are two sub-protocols used to communicate between systems. The communication protocol between a DUA (Client) and a DSA (Server) is called the Directory Access Protocol (DAP). The communication protocol between one DSA (Server) and another DSA is called the Directory System Protocol (DSP). The X.500 uses the DSP sub-protocol to give a "distributed" and "global view" of the data. That is, not all the data is stored on one server but distributed among multiple servers. However, when a client accesses a X.500 system via DAP, the data is gathered from one or several servers using DSP and presented as one global view of the data.

2.1 Directory Services

What is a directory in computing terms? A classic analogy is the white and yellow pages of a telephone book. A common feature of both white and yellow pages is the ability to search for information; the difference in the two is the way they are indexed.

Publishing information in a directory and allowing users, applications, and systems administrators to make use of this information is the fundamental advantage of a Directory.

Directories, such as Lightweight Directory Access Protocol (LDAP) and Active Directory (AD) are types of databases that can be searched to provide useful network information. A user can find network information without any knowledge of the structure of the network. For example, the user can search the Active Directory for a share, requiring no knowledge of the network. This is because the directory has abstracted a server's share to a directory share. Without Directory Services, a user has to know the server name and its share name to mount a network file share. AD changes this.

Searching is a fundamental service provided by LDAP, so the more information "published" in the directory, the more productive the user community becomes. LDAP is a standard and the Active Directory is LDAP compliant. Since AD adheres to the LDAP standard, third party applications are leveraging the directory. AD-aware applications can use Windows 2000 services for authentication and access controls. These applications can store configuration information in the directory

"NOS" is the term used to describe a networked environment in which various types of resources, such as user, group, and computer accounts, are stored in a central repository that is controlled and accessible to end users. Typically a NOS environment is comprised of one or more servers that provide NOS services, such as authentication and account manipulation, and multiple end users that access those services. Microsoft's first integrated NOS environment became available in 1990 with the release of Windows NT 3.0, which combined many features of the LAN Manager protocols and OS/2 operating system. The NT NOS slowly evolved over the next eight years until Active Directory was first released in beta in 1997. Under Windows NT, the "domain" concept was introduced, providing a way to group

resources based on administrative and security boundaries. NT domains are flat structures limited to about 40,000 objects (users, groups, and computers). For large organizations, this limitation imposed superficial boundaries on the design of the domain structure. Often, domains were geographically limited as well because the replication of data between domain controllers (i.e., servers providing the NOS services to end users) performed poorly over high-latency or low bandwidth links. Another significant problem with the NT NOS was delegation of administration, which typically tended to be an all-or-nothing matter at the domain level. Microsoft was well aware of these limitations and needed to re architect their NOS model into something that would be much more scalable and flexible. For that reason, they looked to LDAP-based directory services as a possible solution.

1.1.1 Brief History of Directories

In generic terms, a directory service is a repository of network, application, or NOS information that is useful to multiple applications or users. Under this definition, the Windows NT NOS is a type of directory service. In fact, there are many different types of directories, including Internet white pages, email systems, and even the Domain Name System (DNS). While each of these systems have characteristics of a directory service, X.500 and the Lightweight Directory Access Protocol (LDAP) define the standards for how a true directory service is implemented and accessed. In 1988, the International Telecommunication Union (ITU) and International Organization of Standardization (ISO) teamed up to develop a series of standards around directory services, which has come to be known as X.500. While X.500 proved to be a good model for structuring a directory and provided a lot of functionality around advanced operations and security, it was difficult to implement clients to utilize it. One reason is that X.500 is based on the OSI (Open System Interconnection) protocol stack instead of TCP/IP, which had become the standard for the Internet. The X.500 directory access protocol (DAP) was very complex and implemented a lot of features most clients never needed. This prevented large-scale adoption. It is for this reason that a group headed by the University of Michigan started work on a "lightweight" X.500 access protocol that would make X.500 easier to utilize.

The first version of the Lightweight Directory Access Protocol (LDAP) was released in 1993 as RFC 1487, but due to the absence of many features provided by X.500, it never really took off. It wasn't until LDAPv2 was released in 1995 as RFC 1777 that LDAP started to gain popularity. Prior to LDAPv2, the primary use of LDAP was as a gateway between X.500 servers. Simplified clients would interface with the LDAP gateway, which would translate the requests and submit it to the X.500 server.

The University of Michigan team thought that if LDAP could provide most of the functionality necessary to most clients, they could remove the middleman (the gateway) and develop an LDAP-enabled directory server. This directory server could use many of the concepts from X.500, including the data model, but would leave out all the overhead provided by the numerous features it implemented. Thus the first LDAP directory server was released in late 1995 by the University of Michigan team, and it turned into the basis for many future directory servers. In 1997, the last major update to the LDAP specification was described in RFC 2251. It provided several new

features and made LDAP robust enough and extensible enough to be suitable for most vendors to implement. Since then, companies such as Netscape, Sun, Novell, and Microsoft have developed LDAP-based directory servers. Most recently, RFC 3377 was released, which summarizes all of the major LDAP RFCs.

2.3.1 Domain

The core unit of logical structure in the Active Directory is the domain, which can store millions of objects. Objects stored in the domain are considered "Interesting" to the network. "Interesting" objects are items the networking

community members need to do their jobs: printers, documents, e-mail addresses, databases, users, and other resources. All network objects exist within a domain and each domain stores information only about objects it contains. Active Directory is made up of one or more domains. Grouping objects into one or more domains will allow the network to reflect a DOE site's organization. Domains will allow each internal division to partition their information from the rest of the organization. Domains share these characteristics:

- All network objects exist within a domain and each domain stores information only about the objects that it contains.
- A domain is a security boundary. Access control lists (ACLs) control access to domain objects. All security policies and settings such as administrative rights, security policies, and ACLs do not cross from one domain to another.
- The Domain Administrator has absolute rights to set policies only within that domain.

The atomic unit of the Windows 2000 is the domain. A domain is an administrative boundary,